

Project Scope: "Pixel Defender" (1-Hour AI-Assisted Web Game)

1. Project Overview

- **Project Name:** Pixel Defender
- **Objective:** Build and deploy a fully functional, browser-based 2D arcade shooter in under 60 minutes using AI coding assistants.
- **Target Audience:** Casual web players.
- **Core Loop:** Control a ship at the bottom of the screen, dodge or shoot descending asteroids/enemies, collect points, and try to beat a high score before losing all 3 lives.

2. Technical Stack

- **Frontend Architecture:** Single-file `index.html` containing HTML5 Canvas, CSS3 styling, and Vanilla JavaScript (ES6+).
- **Game Engine:** HTML5 Canvas API (no external libraries to minimize setup overhead).
- **AI Tooling:** Claude 3.5 Sonnet, ChatGPT Plus (GPT-4o), or Cursor IDE.

3. Feature Breakdown

In-Scope (Must Have for 60-Minute Build)

- **Player Movement:** Left/Right arrow keys or `A/D` keys to position the ship; `Spacebar` to shoot.
- **Enemies:** Basic enemy shapes spawning at random X-coordinates from the top, moving downward at increasing speed over time.
- **Collision Detection:** AABB (Axis-Aligned Bounding Box) or radius-based collision for laser-to-enemy and enemy-to-player hits.
- **Scoring & UI:** Real-time Score Counter, Level/Speed Escalation, and a 3-Heart Health System overlaid on screen.
- **Game States:** Start Screen (Press Space to Begin), Game Loop, Game Over Screen with a "Restart" button.
- **Visual FX:** Basic particle explosions (Canvas-drawn circles that scatter and fade on enemy destruction).